SCIENCE & INNOVATION FOR SUSTAINABILITY

KRISHI-DRISHTI: TECHNICAL EVIDENCE & SUPPORTING ANNEXURE

Document 03: Technical Blueprints, Circuit Schematics, Scientific Formulations & Test Data

Why we created this Technical Notebook: *When you build an electronics and AI project, you cannot squeeze every circuit wire, every mathematical filter, and every raw test table into the main report because of the strict 3,500-word limit. My partner Rajeev and I prepared this Technical Annexure as our project's scientific evidence file. Here, we document our exact pinout connections, the math we programmed to filter out blurry photos, our dataset deduplication audit, all our benchmark test numbers, and a simple glossary of technical terms we learned along the way.*

SECTION 1: HARDWARE WIRING, INTERFACES & PINOUT TABLE

We spent weeks testing jumper wires on breadboards. Here is our exact wiring connection table, verified directly from our compiled Arduino code files (KRISHI_DRISHTI_Controller.ino and KRISHI_DRISHTI_Camera.ino):

Subsystem	Component Pin	ESP32 Microcontroller GPIO	Signal Type / Electrical Specification	Firmware Configuration / Pull-up
Controller Wand	SSD1306 SDA	GPIO 21	I2C Serial Data (3.3V)	Internal Pullup / 400kHz Fast Mode
Controller Wand	SSD1306 SCL	GPIO 22	I2C Serial Clock (3.3V)	Internal Pullup / 400kHz Fast Mode
Controller Wand	SCAN Button	GPIO 27	Active LOW momentary switch	INPUT_PULLUP, software debounced 50ms
Controller Wand	NEXT Button	GPIO 26	Active LOW momentary switch	INPUT_PULLUP, software debounced 50ms
Controller Wand	DHT11 DATA	GPIO 18	Single-wire bi-directional digital	Bit-banged timing, 10k external pullup
Controller Wand	Power In (VCC)	5V / VIN pin	Regulated 5.0V DC from power bank	Filtered via 100uF capacitor
Camera Unit	Flash Torch LED	GPIO 4	High-power onboard white LED	PWM controlled / active HIGH trigger
Camera Unit	OV5640 SIOD (SDA)	GPIO 26	SCCB Serial Data interface	Hardware I2C bridge to camera sensor
Camera Unit	OV5640 SIOC (SCL)	GPIO 27	SCCB Serial Clock interface	Hardware I2C bridge to camera sensor
Camera Unit	OV5640 XCLK	GPIO 0	Master Clock output (20 MHz)	LEDC hardware timer square wave
Camera Unit	MicroSD D0 / CLK	GPIO 2 / GPIO 14	SD_MMC 1-Bit high speed bus	DMA accelerated block transfer (512B)

SECTION 2: HAND-DRAWABLE CIRCUIT SCHEMATIC (ASCII LAYOUT)

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| | | | | | |
| [OLED SSD1306] [DHT11 SENSOR] [SCAN BTN] [NEXT BTN] |
| VCC -> 3.3V VCC -> 3.3V PIN -> 27 PIN -> 26 |
| GND -> GND GND -> GND GND -> GND GND -> GND |
| SDA -> GPIO 21 DATA -> GPIO 18 |
| SCL -> GPIO 22 (10k pullup to 3.3V) |
+-------------------------------------------------------------------------------+
|| Wi-Fi 802.11 b/g/n (SoftAP @ 192.168.4.1)
+-------------------------------------------------------------------------------+
| AI-THINKER ESP32-CAM UNIT |
| 5V In <-- Power Bank | GND <-- Common GND |
| OV5640 5MP Sensor <-- DVP 8-bit bus (GPIOs 35,34,39,36,21,19,18,5) |
| Flash LED <-- GPIO 4 (Diffuse acrylic disc) |
| MicroSD Card <-- SD_MMC 1-Bit mode (GPIOs 2, 14, 15) |
| Embedded PROGMEM HTTP Server: Port 80 (Endpoints: /api/capture, /api/diagnosis)|
+-------------------------------------------------------------------------------+
```

SECTION 3: HOW OUR BOARDS TALK OVER WI-FI (WITHOUT INTERNET)

People often ask us: 'If there is no 4G signal in the field, how do your boards send photos?' The secret is that our ESP32-CAM creates its own offline Wi-Fi hotspot (SSID: KRISHI-DRISHTI, IP: 192.168.4.1). It acts like a tiny local router in the farmer's hand. No cellular data or SIM card is ever needed. Here are the five simple web requests we programmed:

API Endpoint	Method	Who Calls Whom	Parameters Sent	What Happens in the System
/api/status	GET	Wand --> Camera	None	Checks if camera and SD card are ready to snap photos.
/api/capture	GET	Wand --> Camera	None	Triggers camera to take UXGA JPEG and save it to the SD card.
/api/telemetry/set	GET	Wand --> Camera	?temp=22.4&hum;=88.5	Stores real-time leaf temperature and humidity into memory.
/api/diagnosis/set	GET	Phone --> Camera	?crop=Tomato&disease;=LB &conf;=99.5	Phone uploads the neural prediction and prescription advice.
/api/diagnosis/latest	GET	Wand --> Camera	None	Wand checks every 250ms until diagnosis is ready, then updates OLED.

SECTION 4: THE MATH WE PROGRAMMED TO REJECT BAD PHOTOS

Commercial apps make wild guesses when a photo is blurry or shows dry mud. We couldn't let our wand do that. So we programmed three mathematical filters into our Python script (krishi_ai_daemon.py):

1. Green and Brown Leaf Check (Foliar Ratio):
We inspect each pixel: is it green (chlorophyll: $g > r$ and $g > b$) or brown/black (necrotic rot: $r > 60, g > 40, b < 60$)?
$$\text{Foliar Ratio} = \frac{\text{Green Pixels} + \text{Necrotic Pixels}}{\text{Total Pixels}} \times 100\%$$
If less than 12.0% of the photo has leaf pixels, the phone immediately rejects it as NON_LEAF (bare soil, fingers, or clothes).

2. Dark Border Check (Void Gate):
We measure the brightness of border pixels around the edges of the photo. If over 45% of border pixels are pitch dark (brightness < 35), it flags a dark void warning so dark studio photos or harsh torch fall-offs are not confused with real outdoor leaves.

3. Confusion Index (Shannon Information Entropy):
How confused is the neural network across its 6 class predictions (p_1 to p_6)?
$$H = -\sum_{i=1}^{6} p_i \ln(p_i) \quad [\text{in nats}]$$
• If the model is 99% confident: $H \approx 0.06$ nats (very low confusion).
• If the model is totally confused (guessing ~16.7% on everything): $H = \ln(6) \approx 1.79$ nats.
• **Our Safe Rule:** If $H > 1.45$ nats OR top confidence is under 50%, the screen says UNCERTAIN / RESCAN instead of giving wrong advice!

SECTION 5: DATASET CLEAN-UP & FINDING UPSTREAM DUPLICATES

We wanted to make sure our model's accuracy was 100% genuine and not based on memorizing test images. We wrote an automated audit script (audit_5_duplicate_hashes.py) that generated SHA-256 digital fingerprints for every single image in our collection. Here is what our cryptographic audit discovered:

• **Source Collection:** 6,600 PlantVillage photos (1,100 per class) + 1,110 clean outdoor PlantDoc photos (370 unique images replicated 3x) = **7,710 total records**.

• **Exact Unique Hashes:** The 7,710 records resolve to exactly **6,965 unique image hashes**.

• **The 5 Duplicate Pairs Investigation:** We investigated why 6,600 PlantVillage files had 6,595 hashes (a difference of 5). Our script proved that the original Kaggle PlantVillage dataset has 5 duplicate photos saved under completely different file names! Crucially, our audit verified that each duplicate pair shares the exact same class label—meaning there were zero conflicting labels:

Duplicate Hash Prefix	Class Category	First File Name in Dataset	Second File Name in Dataset	Our Audit Finding
90220c67...	Tomato Late Blight	348c5e63-9562-4ef8...JPG	d1ea0e62-c0cf-46d2...JPG	Identical label (Zero conflict)
1d202e8c...	Tomato Late Blight	578e93aa-c0f5-46aa...JPG	672e811f-c081-432d...JPG	Identical label (Zero conflict)
7ff8410d...	Tomato Late Blight	7f153a77-33f7-41e1...JPG	b1e102f9-723a-4eb3...JPG	Identical label (Zero conflict)
6e1610d4...	Tomato Late Blight	8f3957eb-30e7-49f8...JPG	81966a01-209a-4c28...JPG	Identical label (Zero conflict)
32e9ffab...	Tomato Healthy	4c0b624b-f288-444a...JPG	a533da56-5fa5-498c...JPG	Identical label (Zero conflict)

Table 5.2: Final Training Dataset Distribution (Zero Leakage)

Index	Class Taxonomy Label	Crop	Pathogen / Botanical Agent	Unique Hashes	Total Records
0	Tomato___Late_blight	Tomato	*Phytophthora infestans* (Oomycete)	1,180	1,368
1	Tomato___Early_blight	Tomato	*Alternaria solani* (Fungus)	1,232	1,518
2	Tomato___healthy	Tomato	Healthy Foliar Tissue	1,166	1,326
3	Potato___Late_blight	Potato	*Phytophthora infestans* (Oomycete)	1,170	1,268
4	Potato___Early_blight	Potato	*Alternaria solani* (Fungus)	1,121	1,280
5	Potato___healthy	Potato	Healthy Foliar Tissue	1,096	950
Total	**All 6 Classes**	**Both Crops**	**Audited Training Corpus**	**6,965**	**7,710**

SECTION 6: OUR ACTUAL BENCHMARK NUMBERS & TEST RESULTS

Table 6.1: Clean Botanical Benchmark (60 Fresh Leaves — 10 per class)

Foliar Category	Leaves Tested	Strict Exact Match	Strict Accuracy	Mean Confidence	Inference Latency
Potato Early Blight	10	9 / 10	90.0%	99.8%	9.33 ms
Potato Late Blight	10	10 / 10	100.0%	96.1%	8.40 ms
Potato Healthy	10	10 / 10	100.0%	96.2%	6.19 ms
Tomato Early Blight	10	7 / 10	70.0%	85.0%	6.27 ms
Tomato Late Blight	10	7 / 10	70.0%	81.3%	8.04 ms
Tomato Healthy	10	10 / 10	100.0%	100.0%	6.93 ms
Overall / Mean	**60**	**53 / 60**	**88.33% Strict**	**93.1%**	**7.53 ms**

**Important Scientific Note on Pathogen Match: 1 Tomato Late Blight leaf was predicted as Potato Late Blight. In our strict 6-class system, this is marked as a mismatch (giving 53/60 = 88.33%), but biologically both are caused by the exact same oomycete (Phytophthora infestans), giving 54/60 (90.00%) pathogen-family accuracy.*

Table 6.2: Selected Error Cases and What Went Wrong

Image File	Actual Leaf	What Model Predicted	Confidence	Why It Failed (Root Cause)	How We Fixed / Mitigated It
healthy_potato_leaves.jpg	Potato Healthy	Potato Late Blight	71.1%	Harsh direct sunlight cast dark shadows on lower leaves.	Use optical shroud to block diagonal sunlight.
internet_tea_leaf_ood_01.jpg	Non-Crop (Tea Leaf)	Potato Late Blight	87.0%	Closed-set model forced alien leaf into 1 of 6 classes.	Added Shannon entropy check; inform farmer.
V261110-191530...jpg	Tomato Healthy (Night)	Tomato Early Blight	68.4%	Unshielded flashlight created a white specular hotspot.	Added acrylic diffusion disc over LED torch.
internet_potato_late_02.jpg	Potato Late Blight	UNCERTAIN	0.0%	Black velvet background tripped dark void gate.	Safe gating prevented model from guessing.

SECTION 7: PLAIN-LANGUAGE GUIDE TO OUR CODE FILES

Instead of printing hundreds of lines of code, here is what each file does in plain words:

1. KRISHI_DRISHTI_Controller.ino (Controller Wand Firmware)
Runs on the ESP32 DevKit V1 board. It initializes the SSD1306 OLED screen over I2C and reads the DHT11 sensor on GPIO 18. When the farmer clicks SCAN (GPIO 27), it reads temperature and humidity, sends an HTTP GET request to the camera over Wi-Fi, and begins polling for results. When the diagnosis arrives, it displays the 4-page carousel without using any freezing delay() calls.

2. KRISHI_DRISHTI_Camera.ino (Camera Unit Firmware)
Runs on the AI-Thinker ESP32-CAM. It creates the offline Wi-Fi SoftAP hotspot at 192.168.4.1. When it receives a /api/capture command, it turns on the diffuse flash LED, takes a high-resolution UXGA 1600x1200 JPEG, saves it to the MicroSD card, and updates its internal status to let the phone know an image is ready.

3. krishi_ai_daemon.py (Pocket Smartphone Edge-AI Engine)
Runs locally on the farmer's Android phone using Termux or Pydroid. It connects to the camera's Wi-Fi, downloads the JPEG, center-crops it to 224x224 pixels, and runs our quantized MobileNetV3 TFLite model. It computes the foliar ratio and Shannon entropy, applies our safety gates, and posts the diagnosis back to the camera in under 15 milliseconds.

4. train_high_acc_model.py (Training and Quantization Pipeline)
Our Python training script. It loads our audited 6,965 unique image hashes, applies data augmentations, trains MobileNetV3-Small with categorical cross-entropy loss, and exports the final model to Float16 TFLite format (2.03 MB) with SHA-256 cryptographic verification.

SECTION 8: TECHNICAL GLOSSARY OF SCIENTIFIC TERMS WE LEARNED

• **Edge-AI:** Running artificial intelligence calculations directly on small local devices (like microcontrollers or phones) instead of sending data to giant cloud servers across the internet.

• **Float16 Quantization:** Shrinking neural network weights from 32-bit floating point numbers to 16-bit numbers. This halved our model size from 4.1 MB to 2.03 MB and doubled inference speed with zero accuracy loss.

• **SoftAP (Software Access Point):** A mode where a Wi-Fi chip acts like a mini wireless router, broadcasting its own network name so nearby devices can talk to each other without needing home broadband or phone towers.

• **Shannon Information Entropy:** A mathematical formula that measures uncertainty. In our project, if the model is torn between several diseases, entropy goes high (H > 1.45 nats), triggering an 'UNCERTAIN' safe warning.

• **Cryptographic Hash (SHA-256):** A unique 64-character digital fingerprint of a file. If even a single pixel in an image changes, its hash changes completely. We used hashes to find duplicate images and prevent data leakage.

• **OOD (Out-of-Distribution):** Input data that the AI was never trained to recognize—like placing a tea leaf or a shoe in front of a model trained only on potatoes and tomatoes.

• **Optical Shroud:** A physical black tube attached to the camera lens that sets a fixed focus distance and blocks blinding external sunlight.